

Interview Questions — Python - Advance python programming

Total Questions: 39

1. What does the open() function do in Python file handling?

Hint: Think about how you access files before reading or writing data.

2. Explain the difference between the file modes 'r', 'w', and 'a' when opening a file in Python.

Hint: One is for reading, one for writing (overwriting), and one for appending.

3. How would you write the string 'Hello, World!' to a new text file named 'greeting.txt' in Python?

4. What is the purpose of the tell() method in file handling?

Hint: It relates to the position of the file pointer.

5. How can you move the file pointer to the beginning of a file after reading some lines?

Hint: There's a method that lets you set the cursor position.

6. Write a code snippet to read a file line-by-line and print each line.

7. What is exception handling in Python, and why is it important?

Hint: Consider what happens when your code encounters errors at runtime.

8. How do you handle multiple exceptions in a single try-except block?

9. Write a try-except block that catches a ZeroDivisionError and prints 'Cannot divide by zero.'

10. How do you create a custom exception in Python?

Hint: It involves subclassing a built-in exception class.

11. What does the 'raise' keyword do in Python exception handling?

12. If you see a long traceback after running your code, how can ChatGPT help you understand and fix the error?

Hint: Think about how you would use an AI tool to interpret error messages.

13. What is a class in Python, and how is it different from an object?

Hint: One is a blueprint, the other is a specific instance.

14. Explain the purpose of the `__init__` method in a Python class.

15. Write a simple Python class named 'Car' with attributes 'brand' and 'year'. Create an object of this class.

16. What is single inheritance in Python? Give an example scenario where you might use it.

17. How does multiple inheritance differ from multilevel inheritance in Python?

Hint: Think about the number of parent classes and the inheritance chain.

18. Given a 'Grandparent', 'Parent', and 'Child' class, how would you implement multilevel inheritance?

19. What is polymorphism in Python? Explain with an example.

20. Does Python support method overloading in the same way as languages like Java? Explain your answer.

Hint: Focus on how Python handles functions with the same name but different parameters.

21. How does method overriding work in Python? Give a code example.

22. What is the purpose of a module in Python?

Hint: Think about code organization and reuse.

23. How do you import a built-in module, such as 'math', and use its 'sqrt' function?

24. Write the steps to create your own Python module and import it into another script.

25. What is the difference between a module and a package in Python?

26. What is the use of the 're' module in Python?

Hint: It's related to working with patterns in strings.

27. How does the `match()` function in the `re` module differ from `search()`?

28. Write a Python code snippet to find all occurrences of the word 'python' in a text using `re.findall()`.

29. What is Tkinter used for in Python?

30. How would you create a basic Tkinter window with a label that says 'Welcome'?

Hint: Think about the sequence: import, create window, add label, mainloop.

31. Describe how you would add a button to a Tkinter window that prints 'Button clicked' when pressed.

32. What are the common layout managers in Tkinter, and how do they differ?

Hint: Focus on pack, grid, and place.

33. How do you connect a Python application to a MySQL database using pymysql?

Hint: Think about the connect function and required parameters.

34. Write a code snippet to insert a new record into a 'students' table using Python and pymysql.

35. Explain how you would fetch all rows from a table in MySQL using Python.

36. What precautions should you take when working with databases in Python to avoid SQL injection?

37. How would you use both regular expressions and file handling together to extract all email addresses from a text file?

Hint: Combine reading from a file and searching with a pattern.

38. If you have a class that manages database connections and you want to ensure it only creates one connection object, which OOP concept would you use and how?

39. How could you use ChatGPT or GitHub Copilot to help you debug a Python script that is not working as expected, and what would you do to make sure the suggestions are safe to use?

Hint: Think about verifying AI-generated code and understanding its logic.
